

Tutorial 10

Delta, Gamma, Pinning, and Market-Maker Hedging

A beginner-safe module for reading option-chain pressure before trading options.

Module purpose

Turn the live ODTE discussion into a structured tutorial on Delta, Gamma, open interest, volume, pinning, and dealer hedging. Students learn to observe option-chain pressure on paper before considering any options strategy.

Ideal length	Audience	Student output
120 minutes	Options-chain students	Paper-only strike map + Greek observation log

Conversation source

Prepared from the April 21, 2026 class transcript. External references were added to convert the live discussion into a student-safe tutorial.

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1. Lesson snapshot

The live lesson moved from reading a 0DTE/1DTE option chain into the first real institutional concept: how Delta, Gamma, open interest, and volume can help explain why price may move toward or away from a strike. The student-facing version should keep the analysis paper-only. The goal is to understand pressure, not to trade contracts.

Conversation idea	Student-ready correction	Teaching guardrail
Delta was discussed as price relation and rough probability.	Delta is the estimated change in option premium for a \$1 move in the underlying. Traders sometimes use it as a rough probability proxy, but it is imperfect.	Teach Delta first as sensitivity. Probability language is secondary.
Gamma was described as speed.	Gamma is the estimated change in Delta for a \$1 move in the underlying. It often becomes most important near expiration and near the money.	The speed or acceleration metaphor helps, but it is not the formal definition.
Pinning was connected to high-volume strikes.	Pinning means the underlying may hover near or close near a strike as expiration approaches. Research shows clustering near strikes, but this is a tendency, not a rule.	Never teach pinning as guaranteed manipulation or a buy/sell signal.
Market makers were described as getting delta neutral.	Market makers and dealers may hedge option exposure by trading the underlying or related products. The direction depends on their net position, customer flow, and whether they are long or short Gamma.	Volume alone does not reveal the dealer book.

Clean class mnemonic

Use A-S first: compare the Asset price to nearby Strikes. Then use the G-O-D check: Gamma, Open interest, and Delta. Add Volume as the live participation layer.

Learning objectives

- Define Delta and Gamma in beginner-safe language.
- Explain why Greeks are estimates, not promises.
- Read one expiration at a time before comparing strikes.
- Use volume, open interest, Delta, and Gamma as context, not as trade commands.
- Explain pinning as a possible expiration effect rather than a guaranteed outcome.
- Complete a paper-only observation log for a 0DTE or 1DTE chain.

2. Delta and Gamma in plain English

Delta answers: if the underlying moves by about \$1, how much should this option premium change, all else equal? Gamma answers: if the underlying moves by about \$1, how much should Delta itself change? The phrase all else equal matters because volatility, time decay, spreads, interest rates, news, and liquidity can change at the same time.

Concept	Plain English	Example language for students
Delta	First-order sensitivity of the option price to the underlying price.	If a call has Delta 0.60, a \$1 rise in the underlying might add about \$0.60 to the option, before other variables move.
Gamma	Sensitivity of Delta to the underlying price.	If Gamma is 0.08, a \$1 rise in the underlying might move Delta from 0.60 to about 0.68, before other variables move.
High Gamma area	A place where Delta can change quickly.	Near-the-money, near-expiration options can behave violently because Delta can change fast.

Concept	Plain English	Example language for students
Delta neutral	A position where small underlying moves have reduced directional effect at that moment.	Neutral does not mean risk-free. Gamma, volatility, and time can still hurt the position.

Student warning

A 0DTE option may look cheap, but cheap premium can hide very fast risk. Short time to expiration means there is little time for a wrong idea to recover.

3. The seven-pass chain read

Pass	Question	Action
1. Underlying	What asset is the chain attached to?	Confirm ticker, exchange, and live asset price. Do not mix GLD, spot gold, and futures as if they are the same instrument.
2. Expiration	Which DTE are we studying?	Choose one expiration first. For beginners, compare 0DTE separately from 1DTE and weekly contracts.
3. A-S check	Where is asset price relative to strikes?	Mark at-the-money, then one or two strikes above and below.
4. Volume	Where is live participation appearing?	Compare call and put volume at the same expiration and nearby strikes.
5. Open interest	Where was prior open positioning concentrated?	Use OI as liquidity/positioning context. It is not simply yesterday's volume.
6. G-O-D check	What do Gamma, OI, and Delta say together?	Look for strikes where Delta/Gamma risk may be sensitive and where OI gives liquidity weight.
7. Chart confirmation	Does the chain line up with a chart level?	Compare strikes to support, resistance, VWAP, session high/low, and news catalysts.

4. Pinning without mythology

Pinning is best taught as an expiration-day behavior: when an optionable stock, ETF, or index future trades near a heavily relevant strike as expiration approaches, hedging, liquidity, and exercise-related flows can sometimes help keep price near that strike. Academic research has found clustering near strikes on expiration dates, but the effect can be weak, temporary, overwhelmed by news, or absent.

What strengthens a pinning case?	What weakens a pinning case?
Large relevant open interest near the money.	Major macro news, earnings, Fed events, tariff headlines, or war news.
Very short time to expiration.	High volatility or a fast trend away from the strike.
Underlying trading close to the strike.	Thin liquidity, wide spreads, or conflicting option flow.
Chart level agrees with the strike.	Volume is large but split across calls/puts or expirations.

Instructor language

Say: This strike may be a magnet or a battleground. Do not say: The market has to close here.

5. Market-maker hedging at beginner level

Market makers provide liquidity by quoting options, but they do not want unmanaged risk from every contract they fill. They may hedge by buying or selling shares, futures, or other options. If dealer hedging is one-sided, it can reinforce or dampen

moves. But students should not pretend they can see the whole dealer book from a retail option chain.

Dealer condition	Typical simplified behavior	Student-safe takeaway
Net long Gamma	May buy dips and sell rallies to rebalance.	Can dampen volatility around a zone, but not always.
Net short Gamma	May sell into declines and buy into rallies.	Can amplify a move, especially when price moves fast.
Balanced book	Less urgent directional hedging.	Large volume may not translate into directional pressure.
Unknown book	Retail observer cannot know the full exposure.	Use the chain as context, not certainty.

6. Student worksheet: paper-only strike map

Choose one liquid ETF or index product. Use one expiration only. Fill the table every 30 minutes for one session. Do not place real option trades from this worksheet.

Time	Underlying price	ATM strike	Highest call volume strike	Highest put volume strike	Highest OI strike	Delta/Gamma note	Chart level note
Open							
Mid-morning							
Lunch							
Power hour							
Close							

Completion standard

A passing student can explain why Delta and Gamma changed during the day without turning the observation into a recommendation.

External source stack for students

The source titles below are clickable in most PDF readers. They were selected because they are official, regulator-backed, exchange-backed, or institutionally useful for student verification.

ID	Source	Why it belongs in the tutorial
S1	Options Industry Council - Understanding Options Greeks optionseducation.org	Defines Delta, Gamma, Theta, Vega, and Rho as theoretical guideposts, not guarantees.
S2	OIC - Volatility and the Greeks optionseducation.org	Clarifies that Gamma measures the sensitivity of Delta to a one-unit change in the underlying.
S3	OCC - Characteristics and Risks of Standardized Options theocc.com	Primary risk document students should know before any live options trading.
S4	Cboe - 0DTE Trading Resources cboe.com	Explains short-term option liquidity and the risk of intraday volatility and limited time to expiration.
S5	Cboe - Evaluating the Market Impact of SPX 0DTE Options cboe.com	Useful caution: large notional or volume alone does not equal dealer hedging pressure.
S6	Ni, Pearson, and Poteshman - Stock Price Clustering on Option Expiration Dates papers.ssrn.com	Academic foundation for the idea that optionable stocks can cluster near strikes at expiration.